Ex.No.-1.1-Design and Draft the given 2D modelling software.

AIM:

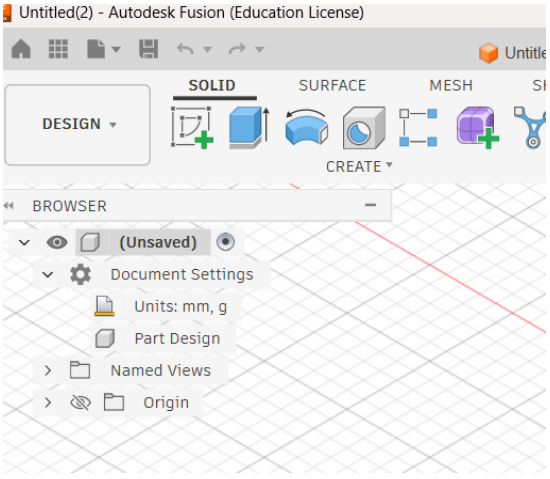
To sketch the given drawing with dimensions using fusion 360 tool.

REQUIREMENTS:

1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 2GB or more)
4. 2.5 Mbps or faster download; 500kbps or faster upload.

PROCEDURE:

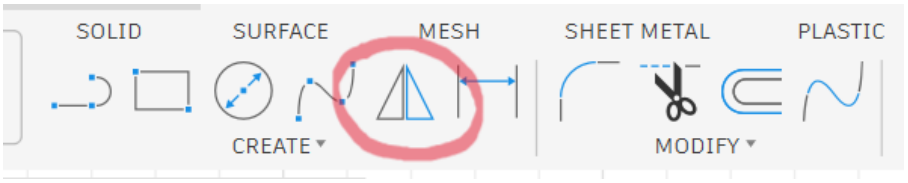
1. Open Autodesk Fusion 360, open a new file, and start a sketch. First, go to Create and select the Front View. Then, start drawing lines and add dimensions.



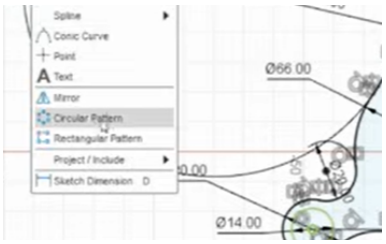
2. Draw a vertical line of 65 mm by selecting the Line tool at the top. Now, draw a circle with a diameter of 40 mm from the centre. With the same centre, draw another circle with a diameter of 66 mm over the 40 mm circle. Also, draw a circle with a diameter of 20 mm and an outer circle with a diameter of 32 mm at the top of the vertical line of 65 mm.
3. From the 65 mm line, draw another two lines of lengths 26 mm and 24 mm vertically and mark the dimensions. Now, draw two lines of 47 mm from the centre. Draw two circles with an inner diameter of 10 mm and

an outer diameter of 20 mm at a vertical distance of 24 mm and a horizontal distance of 47 mm from the centre.

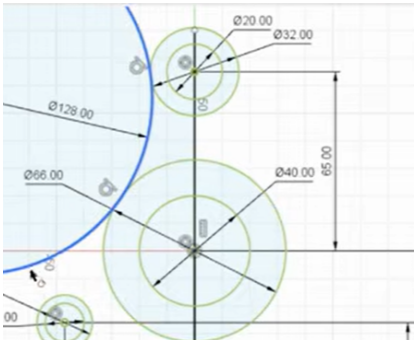
4. Now, draw two circles with an inner radius of 7 mm and an outer radius of 10 mm on either side, at a vertical distance of 26 mm and a horizontal distance of 47 mm from the centre.
5. Using the Mirror tool, select the objects, i.e., the circles of radii 7 mm and 10 mm, and select the mirror line, i.e., the vertical line. By clicking OK, the objects will be projected onto the other side.



6. Now, draw a circle of radius 6 mm on the 40 mm circle using the Circular Pattern tool. Select the object and specify the number of objects to be patterned.
7. After creating six circles of radius 6 mm using the Pattern tool, modify the structure by trimming the top parts of those six circles using the Trim tool.

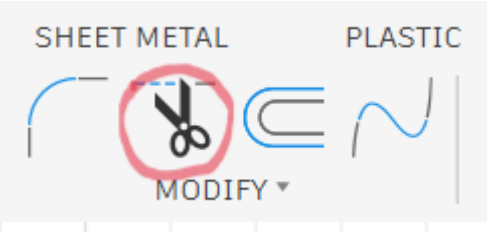


8. Now, draw a circle of radius 64 mm. Open the Constraints menu and select the Tangent tool. First, select the 64 mm radius circle and then select the circle at the top with a diameter of 32 mm. Again, select the 64 mm radius circle and then select the circle with a diameter of 66 mm.

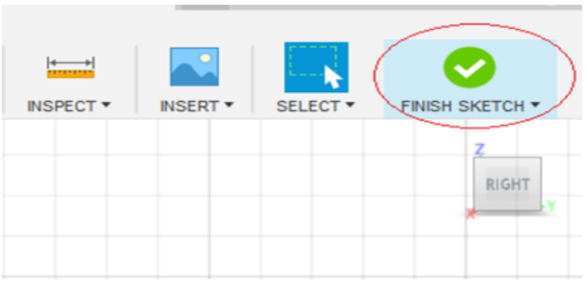


9. Trim the remaining portions using the Trim tool and follow the same procedure for the other side.

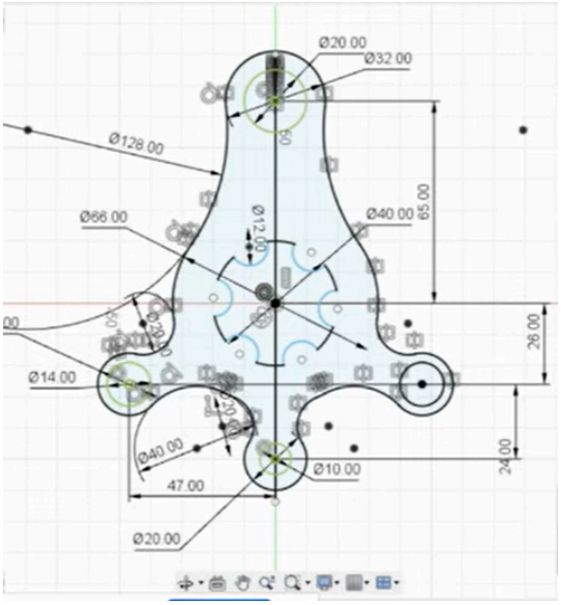
10. Repeat the same procedure by drawing a circle of radius 20 mm. Using the Constraints menu, select the circles and then trim the remaining portions. Do exactly the same for the other side of the sketch.



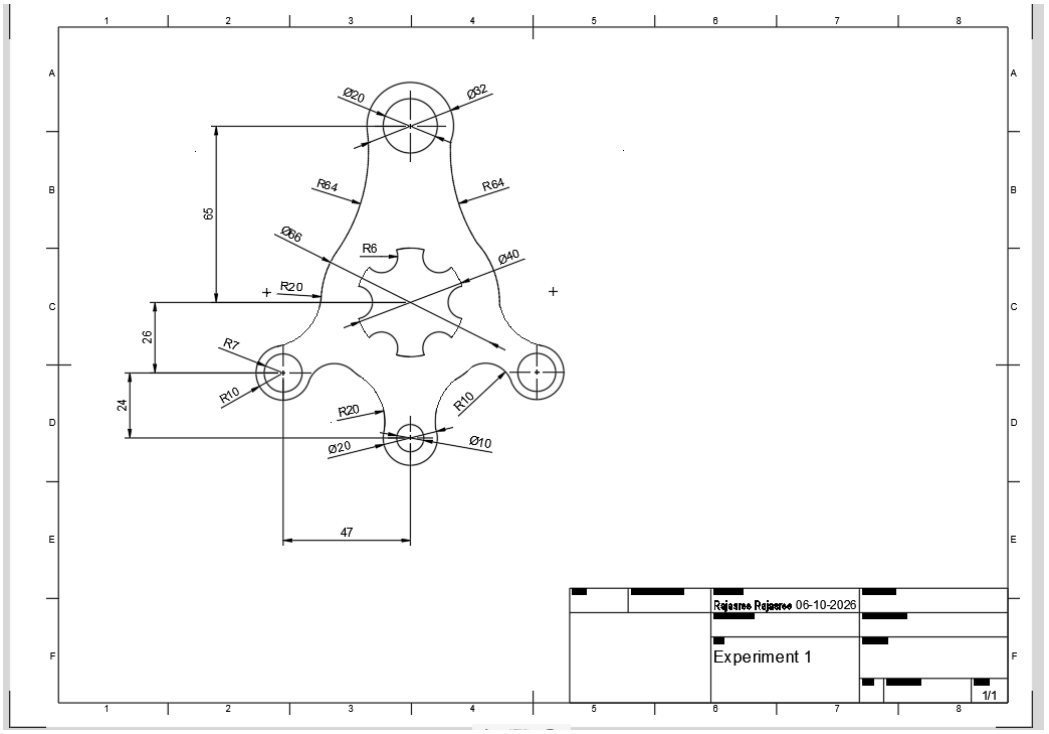
11. Now, draw two circles with radii of 20 mm and 10 mm. Using the Tangent constraint, connect those two circles to the main circle of diameter 66 mm and the circles at the bottom. Using the Mirror tool, you can create the same geometry on the other side.
12. After trimming, click Finish Sketch at the top. After finishing the sketch, switch to Drawing mode to plot the sketch according to ISO.



SKETCH:



OUTPUT:



RESULT:

Thus, the given sketch is drawn and drafted using fusion 360 tool